Hand Gesture Recognition Using MediaPipe and TensorFlow

Mini Project Report

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Abstract

This project presents a real-time hand gesture recognition system using MediaPipe for hand landmark detection and TensorFlow for gesture classification. The solution extracts 21 hand landmarks (63 features) and predicts predefined gestures efficiently.

Objectives

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Recognize hand gestures accurately
Use MediaPipe landmarks
Train a TensorFlow model
Support real-time prediction

Methodology

Workflow:

Dataset Collection → Image Preprocessing → MediaPipe Landmark Extraction → Model Training → Testing → Real-Time Prediction

Technologies Used

• Python

• Google Colab

• OpenCV

• MediaPipe

• TensorFlow

• NumPy

• Matplotlib

Results and Conclusion

The trained model identifies supported hand gestures with high accuracy when evaluated on the test dataset. MediaPipe enables fast landmark extraction, making the system suitable for real-time applications such as touchless interfaces, education, gaming, and sign-language assistance.

Future Scope

Future improvements include adding more gesture classes, supporting dynamic gestures, improving robustness under varying lighting conditions, and deploying the model on mobile and embedded devices.